

Name: _____

Date: _____

PROBABILITY, (MUTUALLY EXCLUSIVE EVENTS)

LO: I can identify mutually exclusive events and use the addition rule.

Mutually exclusive events

Mutually exclusive events cannot happen at the same time. If one occurs, the other **cannot** occur. For example, rolling a 3 and rolling a 5 on one die.

Addition rule: If A and B are mutually exclusive, $P(A \text{ or } B) = P(A) + P(B)$. No need to subtract an intersection because it is zero.

Quick examples

●	$P(2 \text{ or } 5 \text{ on a die}) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$	mutually exclusive
●	Rolling odd and rolling even are ME	cannot be both
●	$P(A \text{ or } B) = P(A) + P(B)$ when ME	addition rule
●	Getting a 3 and getting odd are ME	3 IS odd, so not ME

Key idea

Mutually exclusive events have no overlap: $P(A \text{ and } B) = 0$.
Use simple addition for $P(A \text{ or } B)$.

$$P(A \text{ or } B) = P(A) + P(B)$$

Only when A and B cannot both happen

MODELLLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – identifying ME events

State whether rolling a 6 and rolling an odd number on a die are mutually exclusive.

6 is even, so it is not odd

They cannot happen together

Yes, they are mutually exclusive

Yes, mutually exclusive

6 and odd numbers share no outcomes.

Example 2 – addition rule

A bag has 3 red, 4 blue, 2 green, 1 yellow marble. Find $P(\text{red or green})$.

$$P(\text{red}) = \frac{3}{10}, P(\text{green}) = \frac{2}{10}$$

Events are ME (a marble cannot be two colours)

$$P(\text{red or green}) = \frac{3}{10} + \frac{2}{10} = \frac{1}{2}$$

$$P(\text{red or green}) = \frac{1}{2}$$

Since a marble is one colour only, add the probabilities.

Example 3 – not mutually exclusive

Are 'rolling greater than 3' and 'rolling even' mutually exclusive?

Greater than 3: {4, 5, 6}

Even: {2, 4, 6}

Overlap: {4, 6} $\neq$ empty, so NOT ME

**Not mutually exclusive
(4 and 6 are in both)**

If they share any outcome, they are not mutually exclusive.

Example 4 – exhaustive events

A card is drawn. Find $P(\text{red or black})$.

$$P(\text{red}) = \frac{1}{2}, P(\text{black}) = \frac{1}{2}$$

Red and black are ME (a card is one colour)

$$P(\text{red or black}) = \frac{1}{2} + \frac{1}{2} = 1$$

**$P(\text{red or black}) = 1$
(certain)**

When ME events cover all outcomes, they are also exhaustive.

YOUR TURN – PRACTICE (deliberate practice)

1. Identifying ME events

State whether each pair is mutually exclusive (Yes/ No) for rolling a die.

- a) Rolling a 2 and rolling a 5
- b) Rolling even and rolling a 4
- c) Rolling odd and rolling greater than 5
- d) Rolling a 1 and rolling greater than 4

2. Addition rule

Use $P(A \text{ or } B) = P(A) + P(B)$ for these ME events.

- a) P(1 or 6) on a die
- b) P(red or blue) from 5 red, 3 blue, 2 green
- c) P(vowel or Z) from 26 letters
- d) P(2 or 3 or 4) on a die

3. Not mutually exclusive

Find $P(A \text{ or } B)$ when the events are NOT mutually exclusive.

- a) P(even or >3) on a die
- b) P(heart or king) from a deck of 52
- c) P(prime or odd) from 1-10
- d) P(multiple of 2 or 3) from 1-12

4. Word problems

Solve each problem.

- a) $P(\text{rain}) = 0.3$, $P(\text{snow}) = 0.1$ (ME). $P(\text{rain or snow})$?
- b) $P(\text{bus}) = 0.45$, $P(\text{train}) = 0.3$, $P(\text{walk}) = 0.25$ (ME). $P(\text{not walk})$?
- c) $P(A) = 0.2$, $P(B) = 0.5$, A and B are ME. $P(\text{neither})$?
- d) A spinner: $P(1) = 0.1$, $P(2) = 0.3$, $P(3) = 0.4$. $P(4)$?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Mutually exclusive events cannot happen at the same time

☐

Correct answer: _____

Reason: _____

b) Rolling a 4 and rolling an even number are mutually exclusive

☐

Correct answer: _____

Reason: _____

c) For ME events, $P(A \text{ or } B) = P(A) + P(B)$ ☐

Correct answer: _____

Reason: _____

d) All probabilities in a sample space must add up to 2 ☐

Correct answer: _____

Reason: _____

e) If $P(A \text{ and } B) = 0$, then A and B are mutually exclusive ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A spinner has 5 colours. $P(\text{red}) = 0.2$, $P(\text{blue}) = 0.15$, $P(\text{green}) = 0.3$, $P(\text{yellow}) = 0.25$. Find $P(\text{purple})$. Then find $P(\text{red or blue or purple})$. Are red and purple mutually exclusive? Explain.

Expression: _____

Simplified: _____

b) A card is drawn from a standard **52-card deck**. Explain why 'drawing a heart' and 'drawing a king' are NOT mutually exclusive. Calculate $P(\text{heart or king})$ using the addition rule for non-ME events.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

